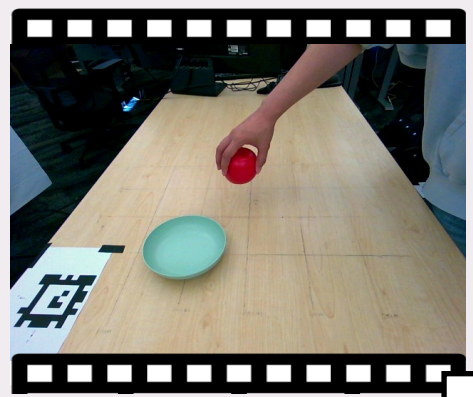


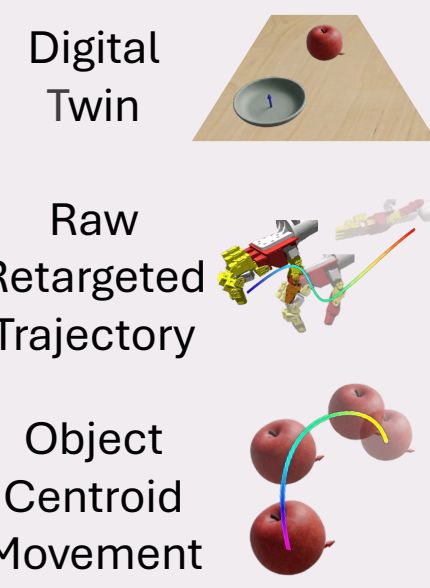
1

REAL-TO-SIM
Off-the-Shelf
Estimation



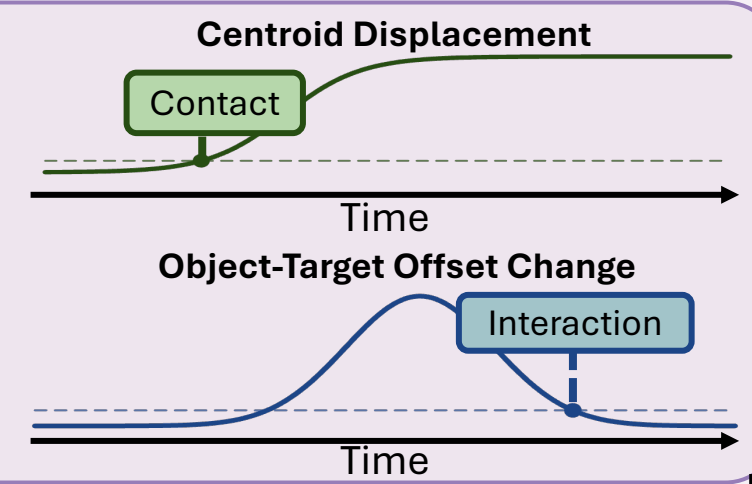
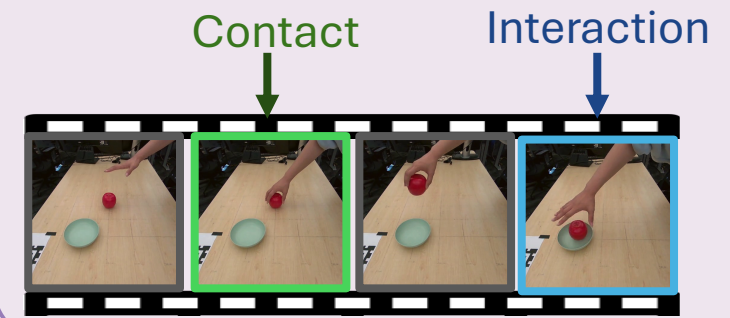
Single Human
RGB-D Video

OUTPUT

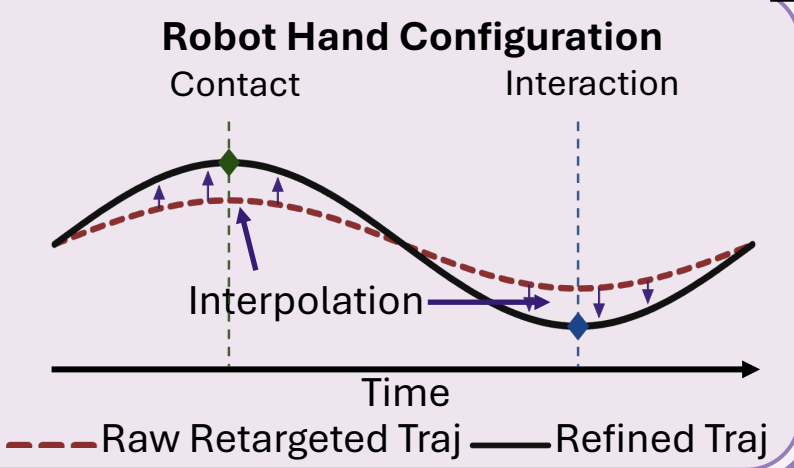
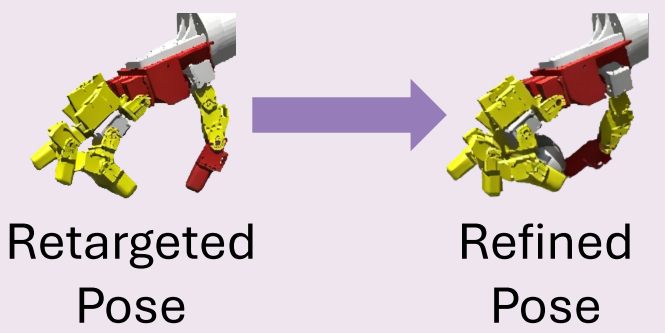


DIGITAL TWIN
Keyframe Refinement

Keyframe Identification



Trajectory Refinement



Refined
Trajectory

OPTIONAL ADD-ON

Collision-Aware Motion Planner

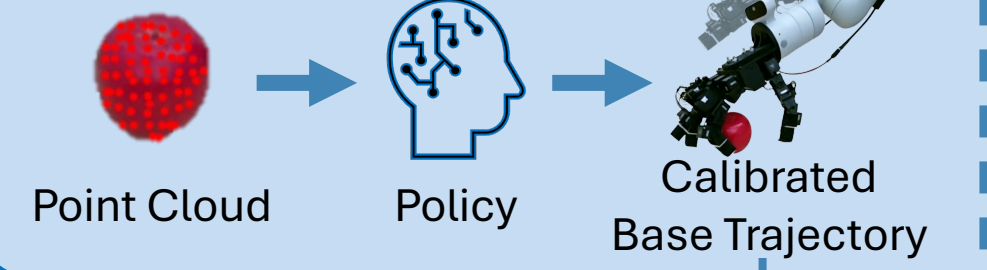
In: refined trajectory + obstacle scene + new task config
Out: collision-free trajectory

Collision-Free
Trajectory

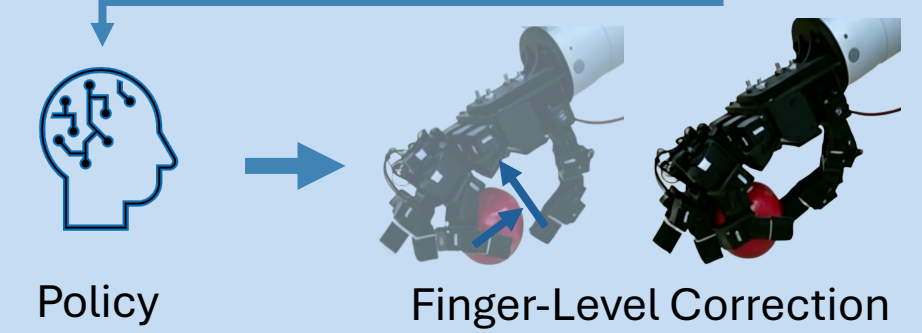
2

SIM-TO-REAL DEPLOYMENT
Decoupled Sim-to-Real Transfer

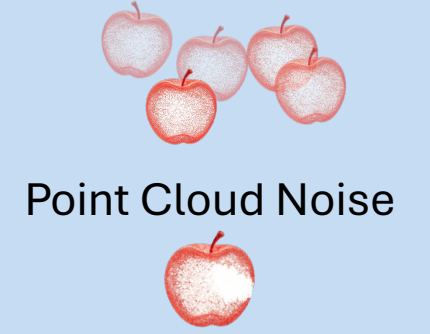
Distillation IL Policy



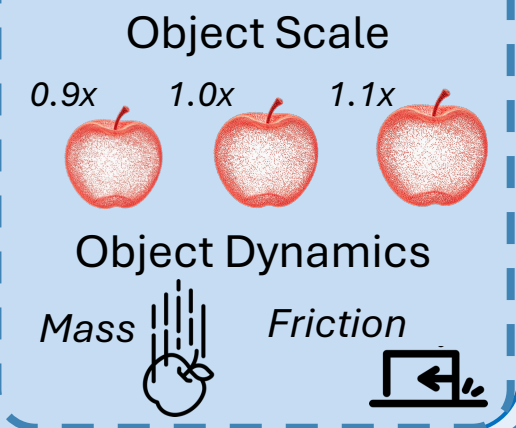
Residual RL Policy



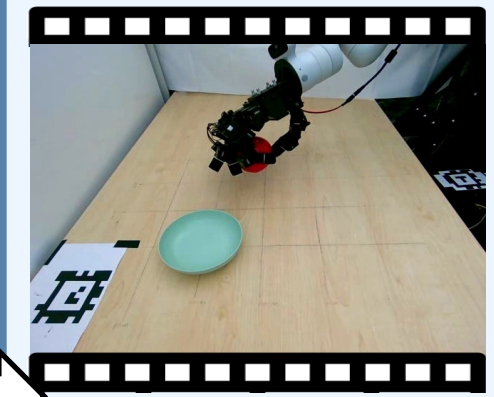
Object Pose



Domain
Randomization



REAL ROBOT
Dexterous
Manipulation



Real-World Task
Execution

OUTPUT

